

Zhenhao Liu

Undergraduate Researcher in NLP

Hangzhou, China
liuzhenhao@westlake.edu.cn



[likc1606.github.io](https://github.com/likc1606) • github.com/LikC1606 • [Westlake NLP Lab](#)

RESEARCH FOCUS

NLP / Language Modeling / Cognitive Alignment / Research Agents

EDUCATION

Westlake University

Hangzhou, China

2024 cohort; GPA 3.753/4.3

Undergraduate, Artificial Intelligence

Selected coursework: Natural Language Processing (A), Machine Learning (A+), Data Structures & Algorithms (A+), Large Language Models (A)

RESEARCH EXPERIENCE

Westlake NLP Lab

Hangzhou, China

[Advisor profile](#)

Undergraduate Researcher; advised by Prof. Yue Zhang

- Lead research on data-efficient Chinese language modeling and cognitive alignment, studying how representation depth and corpus-derived features affect word- and sentence-level fMRI encoding.
- Design reproducible analyses with controlled shared-representation comparisons, participant-cluster bootstrap, leave-one analyses, and sensitivity checks to separate task-specific effects from selection uncertainty.

PUBLICATION

RepDuet at NLPCC 2026 Chinese BabyLM: Task-Specific Hybrid Representations for the Cognitive Track

[Model](#)

Sole first author; accepted to NLPCC 2026 Shared Tasks Poster

[Paper PDF](#)

- Developed RepDuet, a task-specific hybrid representation that combines intermediate contextual states with corpus statistics for word-level evaluation and upper-layer contextual states with lexical and confidence features for sentence-level evaluation.
- Ranked first in the eligible Cognitive Track after organizer reproduction and final-layer verification, achieving 0.33965 on the hidden evaluation; controlled development analyses found a positive margin over the strongest shared representation.

RESEARCH INTERESTS

- **Cognitive NLP and language modeling:** data-efficient Chinese language models, representation learning, and neural encoding.
- **Research agents:** language-model agents for literature search, information synthesis, and research workflows.
- **AI for science:** AI systems that support scientific discovery and the everyday work of researchers.

HONORS

2026	ICPC Asia East Continent Final	Silver Medal
2025	ICPC Asia Xi'an Regional	Gold Medal
2025	CCPC Zhengzhou Site	Gold Medal
2025	ICPC Asia Shenyang Regional	Silver Medal

TECHNICAL SKILLS

Programming Python, Git, testing, and reproducibility
AI/NLP language modeling, cognitive NLP, LLM applications, research agents, and AI for science
Engineering Flask, LangChain, REST APIs, Pytest, Ruff, Mypy, GitHub Actions, and LaTeX